Zongzhen (Jack) Yang

Software Engineer, Real-Time AI Agents and Multimodal Systems

Seattle, WA | jackyangzzh@gmail.com | jackyang.io | linkedin.com/in/jackyangzzh | github.com/jackyangzzh

SUMMARY

Software engineer with 7+ years shipping real-time, multimodal products, now building core systems for the Microsoft 365 Copilot agent in live Teams meetings. Focus: orchestration, tool execution, and context management. Previously engineered the shared UI framework behind Microsoft Mesh for millions of users. M.S. in CS (HCI) in progress at Georgia Tech.

WORK EXPERIENCE

Microsoft — Redmond, WA

Software Engineer II, Teams AI — September 2024 – Present

- Build core systems for the Microsoft 365 Copilot agent (formerly Facilitator) in live, multi-party Teams meetings, shipping real-time notes, agenda tracking, decision capture, and Q&A to customers.
- Design the turn-taking layer of a proactive agent: when and how it speaks in a live group conversation without interrupting.
- Implement the orchestration, tool execution, context management, and multimodal LLM features that let the agent keep up with a live conversation, plus observability and evals.

Software Engineer II, Mixed Reality (Microsoft Mesh, now Teams Immersive) — June 2022 – September 2024

- Engineered UXTools, the shared UI framework behind Microsoft Mesh and Avatars in Teams for millions of users, adopted by every Mesh product team; reusable, accessible components for touch, pointer, keyboard, hand, and controller input.
- Built the main avatar editor for bodies, faces, hair, outfits, and accessories, shipping weekly in 20+ languages. Satya Nadella showed my outfit-selection flow at Microsoft Ignite.
- Key contributor to the avatar rendering and performance pipeline across HoloLens, Quest, web, and desktop; profiled CPU and GPU frame times and applied LOD switching and GPU instancing to maintain target framerates on constrained hardware.

Holos Inc. | AR/VR Interaction Engineer — February 2019 – May 2022

- Developed the multiplayer backend and Unity hand interaction (Leap Motion gesture recognition, object constraints, physics) for a networked AR/VR training platform used by enterprise and government customers.
- Worked with leadership to help secure a $750K U.S. Air Force SBIR Phase II contract and a place in Techstars.

PROJECTS

FakeHaul | fakehaul.com | **Creator & Developer (Next.js, TypeScript, Azure OpenAI)** — July 2026 – Present

- Launched a $0.00-checkout mock store with about 1,800 AI-generated products, peaking at 3K daily active users and growing from 828 monthly visitors in July to 14,375 in September; covered by TODAY, The New York Times, and Fast Company.
- Built an orchestrator for a multi-agent software factory, coordinating coding agents, browser and shell tools, automated tests, and reusable skills; surfaced code diffs and rendered outputs for review.
- Shipped offline AI pipelines to generate, validate, and publish listings, images, and 15,104 reviews, with no runtime model calls.

RESEARCH

Georgia Institute of Technology | HCI Researcher — January 2026 – Present

- Built Ground Tracker, a transcript-analysis system that separates language-model perception from deterministic grounding rules, and a 264-dialogue controlled benchmark; manuscript under consideration at Dialogue & Discourse.

UW Graphics Group, University of Wisconsin–Madison | Graphics Researcher — September 2020 – May 2022

- Built a real-time VR control loop mapping hand motion to a UR5 arm (ROS, RelaxedIK), plus a Unity replay tool for 15+ datasets.

Kats Research Group, University of Wisconsin–Madison | XR Researcher — October 2017 – May 2022

- Developed a computer vision model that simulates color vision deficiency in hyperspectral images with 90% accuracy and ran it in a color-calibrated Oculus Rift demo; co-authored an arXiv publication and a 2021 Journal of Optics paper.

EDUCATION

Georgia Institute of Technology, M.S. in Computer Science (Human-Computer Interaction) — Expected May 2027

University of Wisconsin–Madison, B.S. in Computer Science and Communication (double major) — June 2020

SKILLS

AI and Agent Systems: LLM Agents, Multi-Agent Orchestration, Conversational AI, Real-Time AI, Tool Calling, Context Management, RAG, MCP, A2A, Agent Skills, Multimodal AI, Observability, Evals, Human-in-the-Loop, Agentic Coding, PyTorch

Software and Cloud: Python, C#, C++, TypeScript, JavaScript, Swift, React, Next.js, Azure, ROS, Git, Accessible UI, Profiling

XR and Graphics: Unity, Unreal, OpenXR, WebXR, visionOS, Three.js, Babylon.js, OpenGL, Vulkan, DirectX, Blender, OpenCV